

Lesson 11: Intellectual Property

45 minutes

Overview

Students are asked to reflect on who owns their creative works from this class, such as their pixel images, before reading an article describing how ownership can become complicated as analog works become digital artifacts. After reading the article, students watch several videos explaining copyright and introducing them to the Creative Commons. Students then re-read the article answering three questions about the benefits, harms, and impacts of current copyright policy. Students use their new understanding of copyright to form an opinion about current copyright policies and create a small poster justifying their opinion with a quote from the article. This lesson is also a scaffold to the larger project that begins after this lesson which includes several tasks that are also a part of this lesson such as, annotating an article, answering questions, and forming an opinion using the article as evidence.

Standards

Full Course Alignment

CSP Conceptual Framework

- **IOC-1** - While computing innovations are typically designed to achieve a specific purpose, they may have unintended consequences.

CSTA K-12 Computer Science Standards (2017)

- **IC** - Impacts of Computing

Agenda

Warm Up (5 minutes)

Activity (30 minutes)

Activity Part 1: Read the Article

Activity Part 2: Copyright Overview

Activity Part 3: Article Re-read

Wrap Up (10 minutes)

Assessment: Check For Understanding

Objectives

Students will be able to:

- Argue if current copyright laws are helping or harming society using evidence from an article
- Explain how copyright and Creative Commons Licenses can be applied to digital works of creativity

Preparation

- Print out **Article - Fortnite Stealing Dance Moves** with highlighters ready to mark the text. If you are unable to print copies, see the Teaching Tip below.
- Check the **"Teacher's Lounge"** forum for verified teachers to find additional strategies or resources shared by fellow teachers
- If you are teaching virtually, consider checking our **Virtual Lesson Modifications**

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the teachers

- **Intellectual Property** - Slides

▼ Make a Copy

For the students

- **Article - Fortnite Stealing Dance Moves**
- **Copyright Overview** - Video
- **Copyright in Practice** - Video

- **Creative Commons Copyright** - Video

Vocabulary

- **Creative Commons** - A collection of public copyright licenses that enable the free distribution of an otherwise copyrighted work, used when an author wants to give people the right to share, use, and build upon a work that they have created
- **Intellectual Property** - A work or invention that is the result of creativity, such as a piece of writing or a design, to which one has rights and for which one may apply for a patent, copyright, trademark, etc.

Teaching Guide

Warm Up (5 minutes)

 **Discuss:** *Imagine you were using some of our pixelation tools to create an image and you posted it online for your friends to see - but, a week later you find out someone took that image and put it on a T-shirt that they're selling for \$10 each. How would you feel in this situation?*

Discussion Goal: This discussion doesn't need to come to a definite conclusion - it should generate more questions than answers. You can complicate this discussion by asking:

- What if the owner gives you credit for the drawing on the T-shirt - does that make it more or less okay?
- If we use the pixelation widget, then our image is just a binary number that's been transformed into an image - can we really 'own' a number?
- What if, instead of this happening a week later, it happened 20 years later. Does that change anything?

Remarks

When you create materials on the computer, you own them - they are your "Intellectual Property". Using materials created by someone else and trying to pass them off as your own is plagiarism when you don't have the creator's permissions. There may even be legal consequences for using these materials!

However, we can see from our discussion that it can be complicated when we talk about ownership of digital information. Today we are going to explore these issues.

Activity (30 minutes)

Activity Part 1: Read the Article (10 minutes)

 **Distribute:** **Article - Fortnite Stealing Dance Moves**

To Print or Not To Print? This lesson is written assuming that you have printed out the article and have it physically available for students to write on, even though it is also possible to have students interact with this text digitally. If students read the article digitally, it is most important that they still follow the active reading strategies outlined in this lesson - highlighting the text, writing in the margins, and summarizing. This may require some additional time & instruction to teach students your preferred digital annotation tools and may require some additional adjustments to some of the later annotation strategies in this lesson.

 **Do This:** Students read the article. After they are finished reading they should mark up the text with the following:

- **Highlight / Underline:** Any information in this article that you want to know more about
- **At The End:** Write a 10-word summary of the article

 Teaching Tip

In order to complete the following lesson and project independently, students may need support with the processes of

- Annotating an article
- Answering questions
- Forming an opinion using the article as evidence

It is especially important that students use marking the text strategies to help them comprehend and synthesize the information from their article, because they will need to do this in the next lesson as well.

 **Prompt:** This article brings up issues around copyright. Based on what you've read and your own experiences, what questions do you have about copyright?

 **Display:** Today's central question: Are our current copyright policies helping society or hurting society?

 Teaching Tip

You may find that a topic about digital copyright is in the news when you are teaching this lesson. It may be appropriate and relevant to use those news events in addition to, or instead of, the article in this lesson plan. Any article you use should:

- Discuss the complexities of owning and sharing digital information
- Explore who benefits or is harmed by policies around copyright

If you find current news stories that also cover these points, you may want to consider using those articles here. This is also a good opportunity to visit the [forum](#) to share with other teachers the articles you found.

Activity Part 2: Copyright Overview (10 minutes)

Remarks

Before we can really discuss this question, we need some additional background information. We are going to watch three videos. Our goal is to better understand the rules of copyright, and to better understand when we can reuse or remix something.

-  Video #1: **Copyright Overview** (2 minutes)
-  Video #2: **Copyright in Practice** (4 minutes)
-  Video #3: **Creative Commons Copyright** (3 minutes)

The videos in this section are sourced from the [Copyright & Creativity for Ethical Digital Citizens Curriculum](#). Credit to the Internet Education Foundation and iKeepSafe for these videos.

Videos are used throughout the curriculum to spark discussions, supplement key concepts with additional explanations and examples, and expose students to the various roles and backgrounds of individuals in computer science.

While interacting with the video, turn on closed captioning so students can also read along as they watch.

To encourage active engagement and reflection, use one or more of the strategies discussed in the [Guide to Curriculum Videos](#).

Activity Part 3: Article Re-read (10 minutes)

Remarks

Now that we better understand the rules & controls of copyright, we're going to re-read this article to see if we can determine if current copyright policies are helping or hurting society

Display the central question again: Are our current copyright policies helping society or hurting society?

Do This: Have students re-read the article in order to answer these questions:

- What was digitized?
- What was the goal or purpose of digitizing this thing?
- Is someone benefiting from this situation? If so, who?
- Is someone being harmed in this situation? If so, who?
- Are these impacts intended or unintended? How do you know?

Students should continue to annotate the article by adding the following symbols:

- Add a + next to sentences that show benefit
- Add a - next to sentences that show harm
- Add a face next to sentences that show impact

Discuss: *Share some of the sentences you annotated. Did everyone identify the same areas?*

Discussion Goal: This discussion continues to generate questions and spark student curiosity based on the article. Ultimately we will present students with a focused question to continue the lesson, but this lets students voice their ideas and concerns with the class. It can be helpful to keep these questions & concerns in mind as the lesson continues and return to them when you can.

Activity Repetition: This aspect of the lesson - reading an article and looking at these questions - is repeated in tomorrow's lesson as well. Students are presented with a different set of articles to examine and must also identify benefits, harms, and impacts. Having them complete this same task today acts as a scaffold to prepare students to read an article with purpose. As you circulate and work with students, offer reading tips and strategies that will help them complete this similar task independently tomorrow.

Wrap Up (10 minutes)

Remarks

You need to take a stand on today's question, using this article to help support your position. We will do this by creating a position poster so we can see how everyone feels about Copyright based on this article.

Do This:

- Distribute paper to each student. Have students fold the paper in half.
- In the top section, complete this sentence stem: "I think copyright can [help / hurt] society because _____"
- In the bottom section, provide a quote from the text that helps justify the sentence you wrote in the top

Circulate: Check in with students and encourage them to use one of their highlighted sentences as evidence for their opinion. As students finish, they hang them in a public space edge-to-edge like a quilt to form a larger tapestry of opinions about copyright with evidence. This class artifact can hang in the classroom as a reference for the next few lessons.

Remarks

Today we learned about Creative Commons, a license that allows you to freely use materials created by others. There are a few other licenses that you may have heard about that also allow access:

- Open Source: programs that are made freely available and may be redistributed and modified.
- Open Access: online research output free of restrictions to access and use

You do have options when you want to ethically use others' materials! Because of these licenses, we have access to a wide variety of digital materials.

To close, when you use these materials, always make sure to cite where you found your sources. Citations come in many forms - it could be a link, or a description of where you found the source. There are formal versions of citations recommended by various organizations, but generally the most important thing to remember when citing a source is to provide as much information as possible about the materials you are using.

Teaching Tip

Creating the Artifact: This is another aspect of the lesson that will be repeated in tomorrow's project - students will create an artifact that will be displayed for their peers to see, and this artifact must also include references to the text they read. Displaying the artifacts from today's lesson acts as a model for what students will be expected to do independently in tomorrow's lesson.

Assessment: Check For Understanding

Check For Understanding Question(s) and solutions can be found in each lesson on Code Studio. These questions can be used for an exit ticket.

Question: How is a Creative Commons license different from a regular copyright?

Question: Now that we understand Copyright, what would need to change in order for the scenario from the warm-up to be okay? As a reminder, here was the scenario from the warm-up:

Imagine you were using some of our pixelation tools to create an image and you posted it online for your friends to see - but, a week later you find out someone took that image and put it on a T-shirt that they're selling for \$10 each.



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